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Linking self-assessment to the external environment

An exploratory study

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Abstract *Self-assessment is increasingly prevalent in many organizations. Although managers perceive self-assessment as internally driven, the well-known link between organizational activities and the external environment suggests that outside forces play a significant role. This investigation explores the external motivators of self-assessment through a field study of 14 organizations. Five factors were found to link the conduct of self-assessment to the external environment: availability of an externally developed or sponsored model, presence of a boundary spanning individual, affiliation with professional and trade associations, pressure from powerful external entities, and potential for external reward or recognition. These findings suggest that self-assessment is driven significantly by forces external to the organization. How these external factors combine to form the context of self-assessment may affect the outcomes of the project.*

Introduction
Self-assessment is the holistic evaluation of organizational processes and performance using relatively little outside assistance (Ford and Evans, 2002). Studies suggest that a large number of organizations have conducted, or are planning to conduct, self-assessment (Link and Scott, 2001; Van der Weile *et al.*, 2000). What motivates organizations to conduct self-assessment? Current research suggests that managers generally perceive internal drivers, such as business improvement and quality program stimulation, as primary drivers of self-assessment (Ritchie and Dale, 2000; Van der Weile and Brown, 1999).

However, given the well-known link between organizational activities and the external environment (Lawrence and Lorsch, 1967; Pfeffer and Salancik, 1978 Thompson, 1967), it is likely that outside forces play a significant role in motivating self-assessment. Empirical observation provides clues of external influence. For example, government agencies and professional associations have led the development and diffusion of popular self-assessment frameworks such the Baldrige and European Quality Award criteria (Easton, 1993). Indeed, it is difficult to imagine the current state of practitioner interest in self-assessment in the absence of such institutional influences.

Purpose
The purpose of this investigation is to explore the external motivators of self-assessment in greater detail. Such a study offers fuller specification of the self-assessment phenomenon's context – an important element in theory building (Whetten, 1989). In addition, the outcomes of a self-assessment program may depend

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on the nature of the initiative's motivation. Coercive pressures, for example, may lead to ceremonial adoption of management practices that produce little organizational improvement (Meyer and Rowan, 1977). Thus, better understanding of the external context should improve our ability to predict program outcomes, and to explain why many self-assessment initiatives produce few substantive results (e.g. Conti, 2001). By studying the behavior and context of organizations that are conducting self-assessment, our intent is to elaborate the external forces that influence the practice.

Research method

The research was carried out in 14 organizations. Organizations were selected largely on the basis of their capability to provide access to their self-assessment related activities, including related interactions with outside entities such as professional and trade associations over an extended period of time. Ten of the organizations were manufacturers while four were from the service sector, including one not-for-profit. Organization size ranged from 100 to 2,000 employees.

Our primary interest was to study the environmental context of these organizations with regard to self-assessment, with an eye towards identifying links to external forces that were influencing the conduct of the practice. While semi-structured interviews were conducted with managers in each organization, we were careful not to rely too heavily on this data source. Research that depends on managers' environmental perceptions may not adequately capture elements of external influence, since managers are often unable or unwilling to identify outside forces working on their organizations (Salancik and Meindl, 1984). Moreover, managers' retrospective accounts are prone to inaccuracies (Golden, 1992). Direct observation of organizational activities and context by our research team therefore constituted an important data source. In addition to viewing the conduct of self-assessment events, we were also able to observe various interactions between managers and relevant outside contacts such as professional associations and consultants. Secondary data, such as organizational documents and industry performance information were also obtained. The variety of data enabled the triangulation of sources and enhanced analytical validity (Denzin, 1978). During the course of our investigation, we collected nearly one thousand pages of notes and documents.

We drew from a number of research methodologies developed by social science scholars for generating theoretically useful output from qualitative data. While Glaser and Strauss's (1967) grounded theory notion provided a useful methodological framework, we found the techniques of Strauss and Corbin (1990) and Miles and Huberman (1994) effective for operationalizing the theoretical insights from our qualitative data in a systematic fashion. Using the data, we undertook the process of "memoing" – the generation of narrative, tabular, and graphical documents meant to extract meaning from the low-level data based and create higher-level theoretical categories, concepts, and relationships that help build theory grounded theory. The structured methods of Miles and Huberman (1994) particularly influenced our approach to memo writing. We then generated propositions that formally expressed the theoretical relationships postulated in the memos. Using the constant comparative method (Glaser and Strauss, 1967), we evaluated a proposition's validity against at least one of the following:

- existing data;
- new data;

- logical argument;
- empirical experience; or
- previous theory.

We kept making theoretical comparisons until the marginal findings of additional comparisons were judged insignificant. We were particularly sensitive to data that illuminated how the external environment might be influencing the organizations' efforts to carry out self-assessment projects.

Our analysis produced five factors linking an organization's conduct of self-assessment to its outside world. These factors are presented in the following sections. The presentation approach is similar to reports of other qualitative studies (e.g. Eisenhardt and Bourgeois, 1988) where analytical findings related to a conceptual factor are discussed along with relevant theoretical support and hypotheses that could be used to test the findings in future investigations.

Externally-sponsored assessment models

A common theme among the sample organizations in our study was the use of self-assessment models and frameworks that were developed externally. All 14 of the organizations were using outside models as the basis for their self-assessments. Most organizations were involved in evaluations of their performance management systems using the Baldrige criteria as their frame of reference. Some organizations employed other frameworks as well, including the ISO 9000 standards to assess quality assurance systems compliance. Others were using a model of change implementation developed by an industrial consortium to evaluate change management effectiveness. Each of these models was affiliated with national or international sponsors such as the US Department of Commerce and the International Organization for Standardization.

Managers may be attracted to models sponsored by external institutions because of their impressive affiliation and perceived validity (Ford and Evans, 2002). At one organization, the top management team decided to use an externally-developed model of performance management even though the chief quality officer had developed his own model and had been using it for internal evaluations. When asked why the team did not choose the internally-developed model, one executive responded that the team thought the self-assessment process and findings would be more credible to both employees and to the board of directors (who encouraged the study) if it was developed by outside experts. Strong external affiliation offers an important source of legitimacy for managers seeking to adopt new organizational practices (Hybels, 1995). These findings suggest the following hypothesis for more rigorous future investigation:

H1. Availability of externally sponsored assessment models and frameworks is positively related to the conduct of self-assessment.

Boundary spanners

Boundary spanners serve as liaisons between organizations and their external environment. They constitute a major organizational resource for information about the outside world (Aldrich and Herker, 1977). Through their control of information, boundary spanners shape the diffusion of new ideas between and within organizations (Schwab *et al.*, 1985). Organizations commonly nominate individuals for boundary spanning roles based on technical competence and personal attributes that will help these people link their organizations to external resources (Tushman and Scanlon, 1981).

Nine of 14 organizations employed individuals who served in a boundary spanning capacity relative to self-assessment. These boundary spanners came from various functional backgrounds (see Table I). Three of these individuals were functional quality managers. Four came from other departmental areas such as manufacturing and marketing; two were chief executives. These people were usually well connected to outside organizations or events, and typically belonged to various professional organizations as active contributors.

Boundary spanners commonly assumed three roles during the adoption of self-assessment. These individuals were often regarded as effective teachers in their organizations. As part of this instructional role, these individuals usually oriented top managers about the assessment process and its benefits. Senior management often approved the assessment project based on this instructional effort. Boundary spanners also served as resource acquirers for the assessment project. For example, boundary spanners often selected outside consultants to assist in the data collection, analysis, and reporting phases of the evaluation. Finally, boundary spanners monitored the project, and often contributed to the report containing the assessment findings.

Of particular interest to us was why these boundary spanners brokered the relationship between self-assessment and their organizations. One boundary spanner cited the potential benefits from the holistic evaluation process as the reason why she decided to introduce her organization to self-assessment, stating:

Titles	Role in organization	Role in self-assessment
Continuous improvement coordinator	Agenda setting for executive committee meetings on organizational improvement	Explaining the assessment process to the executive team
Vice president of operations	Train teams on improvement methodology	Managing assessment data gathering and analysis
Quality systems manager	Monitor team performance	Organizing and running meetings discuss findings among executives
Training coordinator	Provide feedback to executive team on organizational improvement progress	Networking with external entities to obtain self-assessment support
Human resources manager	Circulating articles about management innovations among senior managers	
Director of marketing	Organization representative to collaborative external networks	
CQI manager	Developing and maintaining organizational performance measurement systems	
Quality director		

Table I.
Characteristics of boundary spanners

> We've never thought about how all of our departments and people work together as a system, or how all of these processes produce results. Using [self-assessment], we can look at the big picture.

Another boundary spanner viewed self-assessment as a means for engaging executives in a much needed discussion of strategic issues, stating:

> [Self-assessment is] an excuse for getting our management team together in the same room to talk about issues that are killing us.

Other boundary spanners cited external themes. Two noted that self-assessments were "catching on" elsewhere and that the organization should try the practice out. One boundary spanner confided:

> For the past couple of years, I've been going to conferences and reading [trade magazines] that show others doing Baldrige-based assessments. I thought this would be a great thing for us to do. Using this program not only helps us improve, but it sends a powerful message to others – like our customers – about our commitment to quality. In a recent meeting with our largest customer, we let them know that we were doing Baldrige-based evaluations. Down the road, maybe I'll be able to present our results at one of those conferences.

Some of these comments are consistent with Ritchie and Dale's (2000, p. 242) observation that some organizations adopt self-assessment because it seems to be the thing to do. However, some boundary spanners may merely be imitating others, since managers are often attracted to fashionable business practices (Abrahamson, 1991). Boundary spanners might also view self-assessment as a means for improving their legitimacy or reputation among professional colleagues. Similar to other administrative processes and resources (e.g. Brown, 1994; Giddens, 1979; Sewell, 1992), self-assessment may serve as an instrument for amassing or displaying power in social interactions. These findings suggest a second hypothesis for future investigation:

H2. The presence of boundary spanners knowledgeable about the practice of self-assessment is positively related to the conduct of self-assessment.

Professional affiliations

All organizations in this study were active in professional and trade associations that promoted the self-assessment concepts among their members. The primary associations in this regard were the American Society for Quality (ASQ), local Chambers of Commerce, and an industrial consortium that promoted managerial development (see Table II). These affiliations influenced the conduct of self-assessment. One way that professional associations transmitted their influence was through the development and dissemination of self-assessment discourse – what was written or spoken about issues related to self-assessment. For example, the ASQ, a large association of quality professionals with over 100,000 worldwide members, serves as a primary administrator of the Baldrige criteria and process. ASQ offers a variety of publications and seminars on the process and outcomes of self-assessment to its membership. Many boundary spanners in this study traced their interest in self-assessment to articles, books, conferences and other discourse offered by ASQ. Said one individual:

Type of support	American Society for Quality	Chamber of Commerce	Industrial Consortium
Discourse	Publisher and distributor of Baldrige criteria	"How-to" consulting on Baldrige-based assessments	Developed change process self-assessment methodology
Legitimation and affiliation	Co-administrator of Baldrige national quality award	National Chamber network; connection to Baldrige national and state award processes	Credible and familiar source of management methodology
Social	Regional and national conferences that address Baldrige self-assessment issues	Business award and group workshops based on Baldrige criteria	Workshops for group sharing of assessment results
Resource control	Access to training materials and organizational benchmarks for self-assessment	Source of training and consultant on conduct of self-assessment	Owner of proprietary self-assessment model and methodology

Table II.
Support provided by various external groups

> Reading about the Baldrige process in [ASQ's monthly member magazine] and hearing people talk about it at [ASQ's annual conference] got me thinking that we better take a look at [Baldrige-based self-assessment].

Management discourse is thought to play an important role in the adoption of new management practices (Abrahamson and Fairchild, 1999). Such discourse may enable the diffusion of new practices across thousands of dissimilar organizations (Abrahamson, 1991). Discourse encourages broad adoption by reinforcing the belief among managers that their organizations would benefit equally from adopting a practice (Strang and Meyer, 1994). In the institutional context, an organization that adopts a practice is legitimized in the eyes of others, which might increase its ability to succeed or at least survive (DiMaggio and Powell, 1983). The articles, books, conferences, and seminars disseminated by the professional and trade associations in our study suggest the influential role of discourse in an organization's decision to adopt self-assessment.

These external affiliations also transmitted influence by the control they exerted over resources necessary for execution of the self-assessment project. Consistent with the findings of Ritchie and Dale (2000), the self-assessing organizations in this study often relied on external personnel to manage a large portion of the assessment process. For example, consultants with experience as Baldrige examiners were often sought for training or for actual conduct of the evaluation. Indeed, it was difficult to imagine how many of these organizations would have executed their self-assessment projects without the resources provided by these external sources. Resource dependence is a primary means for external entities to exert influence over organizations (Pfeffer and Salancik, 1978). Organizations must acquire resources such as information and guidance to effectively adopt innovations (Ettlie, 1976; Johnson and Chang, 2000). Consortia and trade associations often provide the conduit for transmitting the resources necessary for innovation diffusion (Albrecht and Ropp, 1984; Czepiel, 1975). Professional associations also exert institutional pressure and influence to facilitate the broad adoption of new management practices (Greenwood *et al.*, 2002). These findings suggest a third hypothesis for future testing:

H3. The more managers interact with professional and trade associations that promote self-assessment, the more likely the conduct of the practice inside the organization.

Outsider pressure

Some organizations initiated self-assessment projects in response to pressure exerted from influential outsiders. Two organizations had received notice from customers that quality system assessment would be a future condition of doing business. "We're just getting a head start on them", acknowledged one division manager about the initiation of the self-assessment program in his organization. The bargaining power of large customers and suppliers has, of course, been a fundamental concept of industrial economics and strategic management (e.g. Porter, 1980).

Demands for self-assessment came from non-market entities as well. A number of organizations in this study were subsidiaries of corporate parents or had reporting relationships with regulatory agencies. One parent had initiated annual Baldrige-based self-assessments at each of its operating divisions. Corporate executives felt that the feedback obtained from the evaluations would provide added perspective into each operation. Furthermore, it had been historically difficult for corporate management to compare the effectiveness of its divisions, since the operations at each division were unique. Thus, self-assessment was seen as a means for generating comparative data on the management systems at these divisions. However, the subsidiaries often viewed self-assessment as a "necessary evil" or a reluctant response to a command from corporate headquarters. "We wouldn't be doing this if we didn't have to" and "this is being forced on us" captures the spirit of the disgruntled comments we sometimes heard. In a few cases it was readily apparent that subsidiary managers would eagerly "turn off" their self-assessment programs if enthusiasm waned at the corporate office.

The self-assessment projects at some organizations were motivated by request rather than coercion. The president of an industrial consortium, of which some organizations were members, had asked member CEOs to consider a self-assessment methodology related to change management that the consortium had recently developed. This was not an unusual request. The consortium had developed a number of management methodologies in the past; ongoing membership in the coalition depended on a member's participation in consortium research and development activities. However, many member CEOs were actively involved in administrative activities of the consortium as board and executive committee members. Many of the CEOs also held the consortium president in high regard. A number of executives confided that they thought the president was one of the brightest people they knew. "He's an icon", noted one executive.

Four of this study's CEOs volunteered to try the consortium's self-assessment methodology in their organizations. "[The consortium president] asked us to try it. It was hard to say no", admitted one CEO. "We're doing this at the request of [the consortium president]", explained another CEO at a top management team meeting to initiate the self-assessment project. In this context, the consortium's newly developed self-assessment methodology resembled a practice in its pre-institutionalization phase of adoption, since the method had few adopters and little was known about the practice

(Tolbert and Zucker, 1996). Institutional support behind a new practice often encourages adoption for legitimacy reasons rather than for efficiency reasons (Meyer and Rowan, 1977). Using this perspective, these four organizations could be seen as adopting self-assessment to comply with institutional norms – in this case to serve as an experimental or promotional tool for the consortium. Institutional pressures exerted by parent organizations on their subsidiaries can shape the adoption of management practices (Kostova and Roth, 2002; Oliver, 1991). Whether driven by command or request, self-assessment is often inspired by the influence of external entities. These findings suggest a fourth hypothesis for future testing:

H4. Pressure to conduct self-assessment from influential outsiders will be positively related to the conduct of practice.

External incentives

Many corporate, local, state, and national award processes have been developed based on popular self-assessment models such as the Baldrige criteria (e.g. Johnson, 2001). Most of the organizations in this study operated in environments that offered incentives for the conduct of self-assessment. An architectural firm was undertaking self-assessment as part of a larger effort to apply for the Baldrige national quality award process. "We needed to do this ourselves first in order to understand the process and see where the gaps are", explained this firm's quality director. Managers in several organizations appeared driven by similar motives, viewing self-assessment as a way to "tune up" for external award contests. While most of these managers thought that the evaluations could improve their operations, they also acknowledged the potential recognition stemming from winning one of these awards. "It could help the brand", admitted the marketing manager of one service firm. External awards as motivators of self-assessment has been noted elsewhere. Wilkes and Dale (1998) found that managers at many small businesses perceived applying for an award as a primary driver of self-assessment.

Some of this study's organizations were vying for quality awards sponsored by corporate parents. Self-assessment results comprised one dimension of the award criteria. Managers in these subsidiaries expressed interest in winning those corporate awards, and significant preparation often preceded the evaluation process. The top management team at one subsidiary had met routinely during the previous year to address issues thought to influence their ability to score well during the assessment. Said one participant, "[The division manager] really wants us to win that award this year. He has an item in the annual goals related to it. We've been working hard to get ready for [the assessment]." Incentives such as awards and prizes constitute another means by which parent corporations can influence subsidiary behavior (e.g. Oliver, 1991). Unlike commands, which imply punitive consequences for non-compliance, influence via incentives facilitates a cooperative, normative process for adopting desired practices. Normative processes cause organizations to adopt the adoption behavior appropriate for an organization's environment in order to achieve legitimacy and success (DiMaggio and Powell, 1983). Corporate parents might prefer to implement new practices among their subsidiaries via normative processes since coercion often results in adoption without subsidiary commitment and internalization (Kostova and Roth, 2002). These findings suggest a fifth hypothesis for future investigation:

H5. Presence of external rewards or recognition obtainable from self-assessment will be positively related to the conduct of practice.

Summary

Our study revealed five factors that connect the conduct of self-assessment to an organization's external environment: availability of an externally developed or sponsored model, presence of boundary spanning individuals, affiliation with professional and trade groups, pressure from powerful external entities, and potential for external reward or recognition. Detailed findings associated with these factors for the 14 organizations in our study are summarized in Table III. At least three of the factors were present in each organization; all five factors were observed in five (36 per cent) of the organizations.

The hypothesized relationships between these factors and the conduct of self-assessment are synthesized into the model appearing in Figure 1. The model could serve as the basis for empirical investigation into the external context of self-assessing organizations.

Discussion

While scholars have often focused on the internal drivers of self-assessment (e.g. Ford, 1999; Ford and Evans, 2001), this study highlights the role of external influence. These findings contrast previous reports that managers perceive internal reasons, such as business improvement, as primary motivators for self-assessment (e.g. Van der Weile and Brown, 1999; Ritchie and Dale, 2000). However, managers may not be able to perceive or assign much importance to external forces. Managers are often unaware of, or unwilling to acknowledge, the role of outside influence on their organizations' behavior (Salancik and Meindl, 1984). A practical implication of this study is that

Organization	Externally sponsored model	Boundary spanner	Professional affiliations	Outsider pressure	External incentives
1	B,I	CEO	ASQ	C	Yes
2	B,I	Mfg Supt	ASQ	C	Yes
3	B,I			C	Yes
4	B		IC	C	Yes
5	B,I			C	Yes
6	B			C	Yes
7	B,I	QC Supervisor	ASQ	C	Yes
8	B,I		IC	C	Yes
9	B,C	Quality director	ASQ,CC,IC	R	Yes
10	C	Cont Imp Coord	CC,IC	R	
11	B,C	HR specialist	IC	C,R	
12	B,C	Mktg Mgr	CC,IC	R	Yes
13	C,I	VP Mfg	ASQ,IC	R	
14	C,I	CEO	IC		

Notes: *Externally developed models*: B=Baldrige, C=Industrial consortium change model, I=ISO 9000; *Professional/Trade Association Affiliations*: ASQ=American Society for Quality, CC=Chamber of Commerce, IC=Industrial consortium; *Nature of outsider pressure*: C=Command from customer or corporate office, R=Request from influential outsider

Table III. Factors present in case organizations

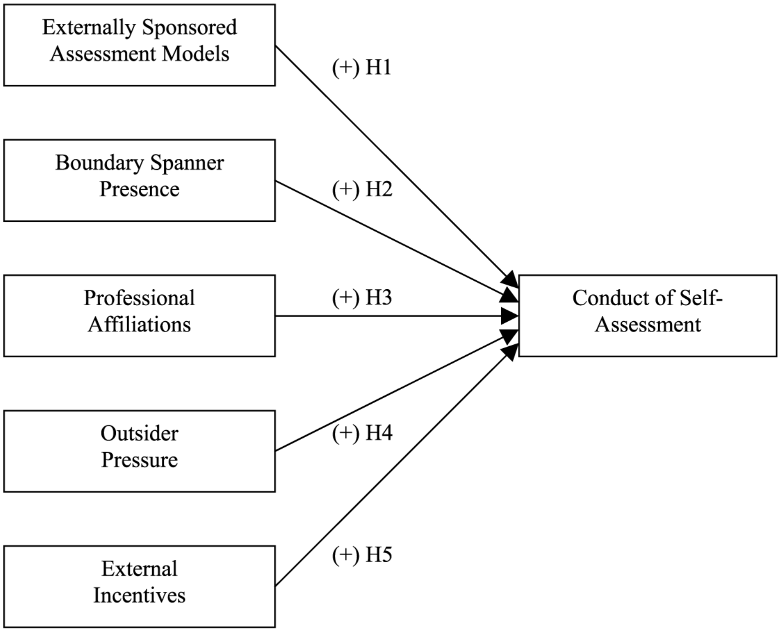

Figure 1.
Model of external influence on self-assessment

managers should be sensitive to this "blind spot" for perceiving external influence and pursue means for increasing their awareness of the external forces working on their self-assessment projects. Such awareness might prevent the conduct of self-assessment for the wrong reasons, and improve the chances of effective outcomes.

As a consequence of our findings, we are prone to consider the "self" in self-assessment a bit of a misnomer. Consistent with a resource dependence perspective (Pfeffer and Salancik, 1978), organizations often appear to rely substantially on outsiders to facilitate self-assessment. It seemed doubtful that some organizations in this study could have executed their self-assessment projects without the training, personnel, and other resources obtained from external providers. In addition to resources, these entities often provide motive. Professional associations, regulatory agencies, and other outsiders use discourse, mandates, incentives, and other means to persuade the conduct of self-assessment. Indeed, institutions constitute an influential force behind the conduct of self-assessment.

The case-based approach of this investigation provides a foundation for larger empirical studies of the contextual factors of self-assessment. Of particular interest would be estimating the relative strength of the relationship between the factors identified in this study and an organization's conduct of self-assessment. For example, do professional and trade group affiliations have more effect on the conduct of self-assessment than, say, outsider pressure or external incentives? Interactions between certain factors are possible. In this study, for example, we noted some relationship between the presence of a boundary spanner and the strength of an organization's interactions with professional and trade associations. Research could also investigate the relationship between the external dimensions identified here and the outcomes of self-assessment. For instance, commands that coerce organizations into self-assessment may lead to undesirable outcomes. Coercion can lead to shallow, ceremonial adoption of a management practice (Meyer and Rowan, 1977). Self-assessment conducted in a ceremonial manner might result in few of the process improvements commonly sought by managers, and could help explain anecdotes of failed assessment projects (e.g., Conti, 2001).

We end by suggesting this study as an example of how operations scholars might integrate the notion of environmental influence into their research. Increasingly, the role of institutions and other environmental actors marks the research agendas of many organizational domains (e.g. Dacin *et al.*, 2002). Operations scholars, however, have been slow to embrace such views in their research. Yet, concepts related to external control and influence might advance research in a variety of operations-related topics. Institutional theory (e.g. DiMaggio and Powell, 1983; Scott, 1995), for example, would seem to offer interesting dimension to scholarship in operations planning (Papke-Shields *et al.*, 2002) and in supply chain power relationships (e.g. Maloni and Benton, 2000; Watson, 2001). We are hopeful that this study encourages others to consider the role of external influence in operations-related inquiry.

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